

Harsh Kumar

Software Engineer — Full Stack — AI/LLM

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SUMMARY

Full-stack developer (React, Node.js, Python, FastAPI) with hands-on experience building **REST APIs**, **real-time systems**, and **AI-powered features** including RAG pipelines, vector search, and LLM integration. Built **3 end-to-end applications** spanning AI document retrieval, certificate automation, and real-time communication. B.Tech Information Technology graduate (**CGPA 8.03**). Completed a **60-hour AI/LLM development internship at HCLTech**.

TECHNICAL SKILLS

Languages	Python, JavaScript, C++, SQL
Frontend/Backend	React, React Native, Node.js, Express.js, FastAPI
Databases	MySQL, MongoDB, Firebase, ChromaDB
AI/ML	RAG, LangChain, Embeddings, Vector Search, LLM Integration, Prompt Engineering
Cloud & Tools	AWS, Microsoft Azure, Git, GitHub, Postman, Vite
Core CS	DSA, OOP, DBMS, Operating Systems, Computer Networks

EDUCATION

2022 – 2026	B.Tech in Information Technology, MIT ADT University, Pune	CGPA: 8.03
	<i>Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Cloud Computing</i>	
2022	Class XII, Kendriya Vidyalaya Southern Command, Pune	85.8%
2020	Class X, Kendriya Vidyalaya Southern Command, Pune	91.2%

EXPERIENCE

HCLTech	Mar 2026 – Apr 2026
<i>AI / LLM Developer Intern</i>	
60-Hour Internship	
Developed LLM-based retrieval workflows using LangChain and ChromaDB to automate document analysis and streamline manual data extraction.	

PROJECTS

DocuMind – RAG-Powered PDF Assistant	 GitHub
<i>React, Python, FastAPI, LangChain, ChromaDB, Hugging Face, Groq, Gemini</i>	

- Designed a FastAPI backend with **7 RESTful endpoints** for PDF processing, semantic search, vector-store inspection, health checks, and LLM-powered chat across **2 LLM providers**: Groq and Gemini.
- Engineered a full-stack RAG application with a conversational React interface for querying **multiple PDF documents** using semantic retrieval.
- Built a **5-stage RAG pipeline** covering PDF extraction, token-based chunking, embeddings, **ChromaDB** vector storage, and **top-3 similarity retrieval**.

Certify – Certificate Generation & Verification Platform	 GitHub
<i>React, Vite, Node.js, Express.js, MySQL, PDFKit, QRCode, XLSX</i>	

- Built a full-stack certificate platform supporting **CSV/Excel uploads**, automated PDF generation, unique certificate IDs, and QR-based public verification.
- Designed REST APIs with **MySQL persistence** for certificate creation, search, downloads, and lifecycle management across **VALID/REVOKED** states.
- Implemented individual PDF and **bulk ZIP downloads**, QR verification, server-side validation, and deployment across **3 services**: Vercel, Railway, and Aiven.

Samvad – Real-Time Assistive Communication App	 GitHub
<i>React, JavaScript, Node.js, Express.js, Socket.IO, Firebase</i>	

- Developed a real-time communication platform with one-to-one messaging, online presence, typing indicators, and live updates using **Socket.IO**.
- Integrated multilingual translation, speech-to-text, and text-to-speech capabilities to support accessible communication across real-time conversations.
- Led a **3-member team** through development and delivery, earning the **Best Departmental Project Award**.

ACHIEVEMENTS

Smart India Hackathon 2024	Top 30 teams in the AI/ML category.
E-Conjecture Hackathon – IIM Indore	Selected as a Finalist .
Best Departmental Project Award	Won for <i>Samvad</i> ; led a 3-member team .

CERTIFICATIONS

Information Technology Specialist in Databases (Certiport) | Python for Data Science (IBM) | AWS Academy Cloud Foundations
Career Essentials in Generative AI (Microsoft & LinkedIn) | Querying Databases with SQL (Coursera)